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AFRAC Guidelines for assessing the appropriateness of few participants in an Interlaboratory comparison programme

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Further information

For further information about this publication, contact the AFRAC Secretariat on:

AFRAC Secretariat
Libertas Office Park,
Cnr Libertas and Highway Streets,
Equestria, Pretoria, 0184
South Africa
Tel: +27 12 740 8539

Email: yolandav@sanas.co.za

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1. Purpose

The purpose of this document is to provide guidelines to accreditation bodies for assessing whether interlaboratory comparisons that have been organised by, and among, only a few laboratories, i.e., 2 to 6 laboratories, including the organiser(s) can be used in the laboratory accreditation process. This document can be used by organisers of and participants in such an Interlaboratory Comparison (ILC) Programme.

Whilst the participation of few participants in an ILC involves mostly two to four participant laboratories, the maximum size of this group is set to six participants, including the organiser(s) of the ILC.

2. Scope

This document is intended to give guidance to accreditation body assessors on which elements of ISO/IEC 17043 are to be taken into consideration when assessing the results of an ILC with few participants (i.e., 2-6 participants), in the frame of laboratory assessments against ISO/IEC 17025 or ISO 15189, and where relevant of inspection bodies against ISO/IEC 17020 (see note).

Note: Proficiency testing may be used in some types of inspection where available and justified by the inclusion of testing activities that directly affect and determine the inspection result or when required by law or by regulators. It is, however, recognized that proficiency testing is not a usual and expected element in the accreditation of most types of inspections."

These guidelines are applicable to ILCs with two to six participants comprising quantitative measurements; similar considerations (but outside the scope of this guidance document) hold for other (e.g., a qualitative) types of ILCs with few participants.

3. Terms and Definitions

Interlaboratory comparison (ILC)
The design, performance and evaluation of measurements or tests on the same or similar items by two or more laboratories in accordance with predetermined conditions" [1]

Interlaboratory comparison with few participants
An interlaboratory comparison organised by, and among six or less laboratories.

Proficiency testing (PT)
Evaluation of participant performance against pre-established criteria by means of interlaboratory comparisons [3]

ILC test item
Sample, product, artefact, reference material, piece of equipment, measurement standard, data set or other information used for proficiency testing [3]

Note: To this document, the ILC test item can be regarded equivalent to the proficiency test item.

ILC organiser
The laboratory which takes responsibility for the development and operation of the ILC [3]

Assigned value
Value attributed to a particular property of a proficiency test item [3]

Note: for the purpose of this document, this is the property value of the ILC test item.

Standard deviation for proficiency assessment *(SDPA, σPT)*
Measure of dispersion used in the evaluation of results of proficiency testing, based on the available information [3]

Reference Material (RM)
Material, sufficiently homogeneous and stable with respect to one or more specified properties, which has been established to be fit for its intended use in a measurement process [8]

Certified reference material (CRM)
Reference material (RM) characterized by a metrologically valid procedure for one or more specified properties, accompanied by a RM certificate that provides the value of the specified property, its associated uncertainty, and a statement of metrological traceability [8]

4. Evaluation of Performance

4.1 The statistical evaluation of participant results for an ILC with few participants is often less straightforward than for an ILC with more than six participants. With a decreasing number of participants, it gets increasingly difficult to identify the distribution of the results, to detect outliers reliably, or to apply robust statistical analysis. It is generally not recommended to derive the assigned value and standard deviation for proficiency assessment (SDPA) from the results obtained by the participants, or at least this should be done with great care and expertise. Similar considerations (but outside the scope of this document) hold for other evaluations, e.g., in case of a qualitative ILC with few participants.

4.2 Accreditation body assessors should give due care to these peculiarities when reviewing the technical relevance and the outcome of an ILC with few participants. To aid this review, the three scenarios below provide examples for a sound evaluation of an ILC with few participants. Which scenario applies in practice depends on the presence and reliability of an externally assigned value, the quality of the dataset, the experience of the participants and the competence and experience of the ILC organiser.

4.3 From a metrological point of view, and in the frame of an ILC with few participants, the use of an assigned value based on an external reference (see Scenario 1 below) should be preferred over an assigned value based on participants' results (see Scenario 2 below), which in turn should be preferred over not using any assigned value (see Scenario 3 below). However, elements from Scenario 3 may also be relevant to the other scenarios because of their educative character.

4.4 In order to establish an evaluation of performance, the ILC organiser should define, pre-assessment criteria, where relevant, before the round is organised.

i) Scenario 1: The organiser has used an assigned value based on an external reference.

The evaluation of results from an ILC with few participants, and performance scoring of participants, are straightforward in this scenario. The organiser may use z scores in which both the assigned value and SDPA are independent of the reported results or use an En number if the assigned value and reported values have stated uncertainties.

The assigned value may stem from a suitable reference standard e.g., the certificate of a CRM or of a measurement standard or instrument in the field of calibration, measurements performed by expert laboratories, or an earlier ILC on the same or a similar material. Similarly, the SDPA could be an external target value that is in line with the results of an earlier ILC or meets specific legislation for which the test has been undertaken. Zeta scores may also be used, preferably in combination with z scores.

ii) Scenario 2: The organiser has used an assigned value based on participants' results.

If an external reference value is not available, quantitative analysis and performance scoring on the basis of the reported results only is generally not recommended. However, there may be exceptions, e.g.:

a) The participants are experienced laboratories that have gathered competence to harmonise their accuracies (trueness and precision) for this measurement, e.g., through earlier rounds of the same or a similar ILC. This is likely to keep the uncertainty of the assigned value small;

b) One of the participants is considered to operate at a higher metrological level (i.e., lower measurement uncertainty), due to the use of reference methodology and more advanced equipment. Its measurement result could be used as the assigned value.

In combination with an external (target) SDPA, the cases a) and b) may be suitable for quantitative analysis and performance scoring.

iii) Scenario 3: The organiser has not used any assigned value.

If no external assigned value is available and an assigned value cannot be reliably calculated from the dataset, then the ILC organiser should not calculate a performance score, however an individual performance may be established. The reported results may for example be graphically displayed and discussed among the small ILC participants. The reproducibility of the results (variation among participants), the repeatability (variation between replicate measurements in the single laboratory under repeatability conditions), the type of distribution, the information contained in extreme values (outliers or not) and the reported measurement uncertainty are examples of information that may be used to establish any individual performance.

5. Assessing Participant Results in an ILC with few participants

5.1 The appropriateness of the participation in an ILC with few participants is to be evaluated when assessing the ILC strategy of the laboratory [4] [5].

5.2 When PT results stem from a PT provider, who operates in accordance with ISO/IEC 17043 are assessed, the focus is mainly on the performance obtained by the laboratory and the criteria used by the PT provider to establish the evaluation of performance. But when assessing the results from an ILC, the operation of the ILC is to be assessed to check that they have been organised in agreement with the relevant requirements of ISO/IEC 17043 and the accreditation body. The assessment depends on which of the following two situations are encountered when assessing a laboratory in relation to an ILC:

- The laboratory assessed has organised and participated in the ILC.
- The laboratory assessed has only participated in the ILC.

5.2.1 In the first situation, the assessor will evaluate the plan in accordance with item 5.2.1.1 and 5.2.1.2 of this document.

5.2.1.1 Planning

A plan, which includes a detailed description of the operation of the ILC with few participants is to be available.

As a minimum, the following points should be included or elaborated in the plan:

- Main contact person;
- If organised jointly, the persons or laboratories involved;
- List of participants;
- The measurand or characteristic to be determined;
- Requirements (production, homogeneity, stability) for the ILC test item
- Information on the use and preparation of the ILC test item (description of the preparation, if applicable);
- Timeframe of the scheme;
- Information on the method(s) to be used;
- Description of the method for the evaluation of the comparability of the results, statistical analysis, if applicable, and the criteria used for the evaluation of performance; and
- Description of the reporting format for the participants and from the organiser.

Preparation of ILC test items
If the organiser prepares the ILC test item itself, then this should be assessed. If not, then all the information relating to the ILC test item e.g., certificates should be checked.

Homogeneity and Stability
Documented evidence of the homogeneity and stability of the ILC test items should be assessed when significant for the evaluation of the small ILC results.

Statistical design
The appropriateness of the statistical design should be assessed.

Assigned value
The assessment should ensure that an appropriate assigned value where relevant and its associated measurement uncertainty is established and treated as "confidential" as possible.

Standard deviation for proficiency assessment (SDPA)
The assessment should ensure that a fit for purpose SDPA has been established.

5.2.1.2 Reports

A report should be established by the ILC organiser. As a minimum, the following points should be included in the report:

- Date of small ILC;
- Contact person;
- Persons or laboratories involved in the organisation of the small ILC;
- Identification of the small ILC scheme;
- Description of the small ILC item;
- The participants results;
- Method for the evaluation of the comparability of the results (assigned value and its associated measurement uncertainty, establishment of the SDPA, range of results, graphical displays);
- Comparability of the participants results and/or participants performance; and
- Comments and recommendations based on the outcome of the small ILC scheme.

Note: If some of the points are clearly included in the plan and the latter is provided to all the participants, then these issues do not need to be included again in the report.

5.2.2 In the second situation, the laboratory should be able to provide details to the assessor on how they have evaluated and decided on the fitness for purpose of the ILC. The assessor should evaluate these details to conclude upon the relevancy of the ILC.

5.3 In addition, it is expected that any unsatisfactory results that are obtained from participation in an ILC are to be treated by the laboratory, like all the other unsatisfactory ILC results, as non-conforming work (see ISO/IEC 17025 and ISO 15189) and the actions taken are to be specifically assessed.

5.4 The criteria used for the evaluation of performance should be fit for purpose.

6. References

[1] ISO/IEC 17025: General requirements for the competence of testing and calibration laboratories.
[2] ISO 15189: Medical laboratories – Requirements for quality and competence.
[3] ISO/IEC 17043: Conformity assessment – General requirements for proficiency testing.
[4] EA-4/18: Guidance on the level and frequency of proficiency testing participation.
[5] EA-4: Guidelines for the assessment of the appropriateness of few laboratory comparison within the process of laboratory accreditation
[6] ILAC-P9: ILAC Policy for Participation in Proficiency Testing Activities.
[7] ILAC-P13: Application of ISO/IEC 17011 for the Accreditation of Proficiency Testing Providers.
[8] ISO Guide 30: Reference material – Selected terms and definitions